Project Planning & Management - Churn Radar

1. Project Proposal & Overview

- **Project Title:** Churn Radar
- **Objective:** To build an end-to-end data analytics and machine learning solution that predicts customer churn in the telecommunications sector, identifies root causes, and provides actionable retention strategies through multi-platform dashboards and a real-time web application.
- **Scope:**
  - Data cleaning and exploratory data analysis (EDA) using Python and SQL.
  - Training predictive classification models (Logistic Regression, Random Forest, XGBoost).
  - Developing executive visualization dashboards (Power BI, Tableau, Looker Studio).
  - Deploying an interactive web application (Streamlit) for real-time risk scoring and automated retention recommendations.

2. Timeline & Milestones

- **Week 1:** Data Cleaning, Preprocessing, and Relational Data Modeling.
- **Week 2:** Exploratory Data Analysis (EDA), Customer Profiling, and Root-Cause Analysis.
- **Week 3:** Machine Learning Model Development, Evaluation, and Retention Strategy Formulation.
- **Week 4:** Multi-Platform Dashboard Development, Streamlit Web App, and Final Executive Presentation.

3. Team Roles & Task Assignment

- **Member 1 (Data Engineer):**
  - Responsible for data preprocessing, handling missing values, data type conversion (e.g., `TotalCharges`), and encoding categorical variables (Week 1).
- **Member 2 (Data Analyst):**
  - Responsible for Exploratory Data Analysis (EDA), customer profiling, and answering core business and churn-driver questions (Week 2).
- **Member 3 (Machine Learning Engineer):**
  - Responsible for building, training, and evaluating classification models (Logistic Regression, Random Forest, XGBoost) and extracting feature importance (Week 3).
- **Member 4 (BI Developer):**
  - Responsible for designing and developing executive visualization dashboards across Power BI, Tableau, and Looker Studio (Week 4).
- **Member 5 (Software & Documentation Lead):**
  - Responsible for system documentation, requirements gathering, database design (ERD), and developing the real-time Streamlit web application (Weeks 1 & 4).

4. Risk Assessment & Mitigation Plan

Risk ID	Risk Description	Impact Level	Mitigation Strategy
Risk 1	**Data Quality Issues** (e.g., missing values, messy data types like TotalCharges).	High	Implement rigorous pandas preprocessing scripts, impute missing values using median/mode, and standardize formats in Week 1.
Risk 2	**Model Overfitting** (ML models failing to generalize well on unseen test data).	Medium	Use cross-validation, evaluate models using robust metrics (ROC-AUC, Precision, Recall), and tune hyperparameters for Random Forest and XGBoost.
Risk 3	**Resource Constraints & Member Inactivity** (Some team members not contributing effectively).	High	Re-distribute tasks evenly across active members with clear ownership of deliverables and modular milestones to ensure smooth execution without delays.

5. Key Performance Indicators (KPIs)

- **Model Performance:** Minimum ROC-AUC score of > 0.80 and high Recall to catch potential churners effectively.
- **Dashboard Usability:** Clean, intuitive UI displaying core KPIs (Churn Rate, Revenue at Risk, Customer Lifetime Value).
- **Timely Delivery:** Adherence to the 4-week structured milestone schedule.